

40.520 Stochastic Models

Spring 2026 Midterm Exam

Guideline

- **Topics:** Necessary Probability Background & Discrete Time Markov Chains
- **Date & Time:** Tuesday, March 3rd, 2:00 – 4:00 PM (Papers handed out at 1:55 pm)
- **Location:** Think Tank 22 (Building 2, Level 3, Room 2.311)
- **Permitted Materials:**
 - A Non-programmable scientific calculator
 - One double-sided A4 cheat sheet
- Scratch paper will be available upon request.
- Applications for the make-up midterm will be accepted until February 27th, 2026.

Review Points

Necessary Probability Background

- Probability axioms and basic rules
- Conditional probability and independence
- Law of Total Probability
- Bayes' Law
- Random variables: distributions, expectation (mean), and variance
- Properties of expectation and variance
- Joint distributions and marginal distributions
- Conditional expectation and the Law of Iterated Expectations
- Central Limit Theorem (CLT)
- Convergence in probability (WLLN) and almost sure convergence (SLLN) (conceptual understanding)
- Time average vs. ensemble average and the Ergodic theorem

Discrete Time Markov Chain

- Markov Property
- Define (discrete-time) Markov Chain, Transition Matrix
- n -Step Probabilities and Chapman–Kolmogorov Equations
- Random Walk and Gambler’s Ruin
- Hitting Probabilities and Hitting Times
- Stopping time
- Communication Classes and Irreducibility
- Recurrence and Transience:
 - Recurrent (Positive Recurrent vs. Null Recurrent)
 - Transient
 - Periodicity
- Stationary Distribution, Long-Run Fraction of Time, and Limiting Probabilities
- Ergodic Theorem: For an irreducible, aperiodic, positive recurrent Markov chain, the limiting probability, stationary distribution, and long-run fraction of time are equivalent.

Important Proofs

One of the following proofs covered in class will appear on the exam to encourage you to review essential material.

- For a nonnegative integer-valued random variable X , $\mathbb{E}[X] = \sum_{n=1}^{\infty} P(X \geq n)$.
- **Wald’s Equation.** Let $S = \sum_{i=1}^N X_i$, where X_i are independent and identically distributed random variables with mean $\mathbb{E}[X]$, and N is a non-negative integer-valued random variable independent of all X_i (for $i = 1, \dots, N$). Then $\mathbb{E}[S] = \mathbb{E}[X] \cdot \mathbb{E}[N]$.
- Use the Markov property to show that the future and past are conditionally independent given the present state.
- **Chapman–Kolmogorov equation.** The probability of transitioning from state i to state j in $n + m$ steps is given by $P_{ij}^{n+m} = \sum_k P_{ik}^n P_{kj}^m$.
- For the gambler’s ruin problem with an infinitely rich adversary, demonstrate that the gambler faces certain ruin when the winning probability per trial satisfies $p \leq \frac{1}{2}$.